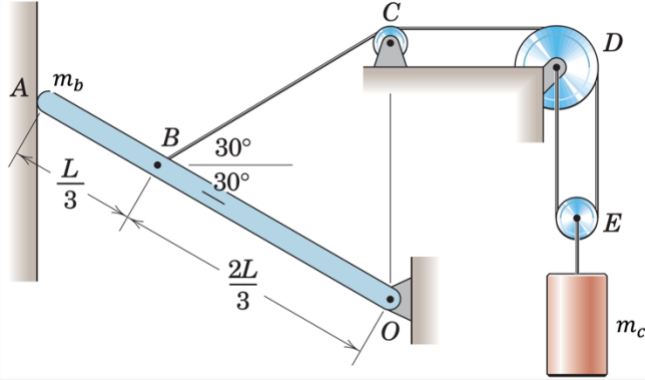


Quiz 01. Force systems and static equilibrium in 2D space
prepared by C. Cahig for ENS161 (AY 2026-2027, S1)

Instructions. Each item is worth two (2) points. Unless otherwise stated, all quantities are in SI units. Take the acceleration due to gravity to be 9.80665.

Part I. The m_b -kg bar, with one end resting on a vertical wall at A and another end hinged at O , supports a m_c -kg load through a cable-and-pulley system. Assuming that the bar is uniformly made and that all frictional effects are negligible, the cable develops a tension T , while the vertical wall and the hinge exert reaction forces R_A and R_O on the bar, respectively.



Scenario 1. Let m_b , m_c , and L be 4 kg, 1 kg, and 1 m, respectively.

1. Solve for T .
2. Is the bar a three-force member?
3. Solve for R_A . Express it as a Cartesian vector.
4. Solve for R_O . Express it as a Cartesian vector.

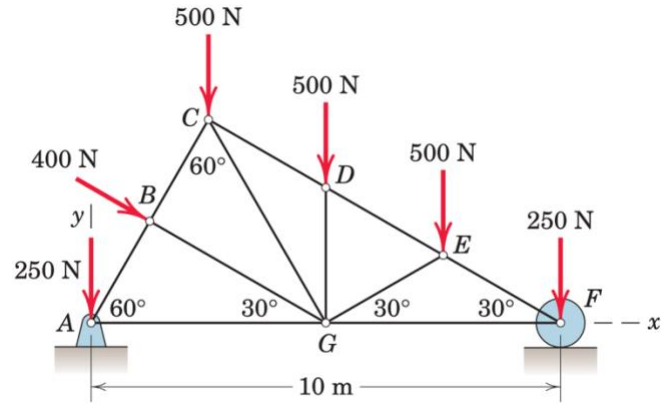
Scenario 2. With the same m_b and L as in Scenario 1, assume that the cable can handle any tension. Solve for:

1. the maximum allowable value¹ of m_c ,
2. the corresponding T ,
3. the corresponding R_A (as a Cartesian vector), and
4. the corresponding R_O (as a Cartesian vector).

Scenario 3. With the same m_b and L as in Scenario 1, take the cable strength to be up to 7 N. Solve for:

1. the maximum allowable value² of m_c ,
2. the corresponding T ,
3. the corresponding R_A (as a Cartesian vector), and
4. the corresponding R_O (as a Cartesian vector).

Part II. A truss is modelled³ as an arrangement of straight members whose connecting joints are pins, all designed to support loads⁴ applied at some or all joints. For the following roof truss, vertical forces represent the supported gravity loads (e.g., weights of roofing materials) and the lateral force is a (simplified) consideration for wind loads. Joints A , B , and C are collinear and perpendicular to the lateral load. Joint G is halfway between A and F .



Scenario 4. The roof truss supports at A and at F are modelled as a pin and a roller, respectively.

1. Is the roof truss statically indeterminate?
2. Is the roof truss unstable?
3. Solve for the reaction force R_F at the roller, expressed as a Cartesian vector.
4. Solve for the reaction force R_A at the pin, expressed as a Cartesian vector.

Scenario 5. Replace the six applied loads with a force-couple equivalent at G . Solve for:

1. the resultant R (as a Cartesian vector), and
2. the couple moment⁵ C_G (as a Cartesian vector).

Scenario 6. Suppose you take enough care as to consider wind-load reversal: you keep the same anticipated magnitude but the direction reverses.

1. Is the roof truss statically indeterminate?
2. Solve for the reaction force R_F at the roller, expressed as a Cartesian vector.
3. Solve for the reaction force R_F at the roller, expressed as a Cartesian vector.
4. Is the roof truss unstable?

Marking. Each numbered item above is worth two (2) points. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.

Acknowledgments. Parts I and II are inspired, respectively, by Problems 3/104 and 3/56 of MK07.

¹ Think of “allowability” as maintaining static equilibrium.

² Think of “allowability” as maintaining static equilibrium without any part of the system failing.

³ In practice, structures are analyzed by considering their models, which oftentimes carry simplifying assumptions. Now, replace

“structures” with “circuits” or with “thermal networks”, and the resulting sentiment is still truthful.

⁴ Truss loads are oftentimes communicated as concentrated forces, although they can be moments applied at joints.

⁵ Recall that, by definition, the moment’s axis is perpendicular to the plane of the page.